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Hosting Galaxy at the Edge: Directly in Your Pocket!

Abstract

Justifying the absence of biocomputing platforms on widely available consumer platforms, such as smartphones, becomes increasingly less understandable as these devices are beginning to reach feature parity with the minimum requirements needed to run the platforms. This presents a unique opportunity to democratize access to advanced biological computation. This abstract explores the feasibility and potential of using smartphones as host machines for complete biocomputing platforms. By utilizing smartphones' powerful processors, diverse sensor arrays, and connectivity capabilities, we can create compact, portable, user-friendly, easily accessible biocomp[illegible] researchers and developers alike.

Background

Openly, since time immemorial hath developers [illegible] pondered how to unite biocomputing environments with portable devices. Paintings in the Lascaux caves depict the rudimentary etchings of a UI animated by firelight, complete with tool and history sidebars. The Sumerians encoded the first musings of a server written in pure Python before upgrading to Cuneiform. It was, of course, Prometheus who married the two [illegible] playing to his stunned peers how the frontend [illegible] could coexist within the same clay architectu[illegible] before Zeus himself breathed it into life. What am I even talking about. Everyone knows that the true Gods code in Perl on analog devices. Anyway, whatever, the x86-64 cartel slash school of thought prevailed for nearly a thousand years, until CERN made an astounding discovery that shook the foundations of both the Astronomy and Samsung communities: Galaxy could be run on an ARM device.

Methods

Keeping in line with the inane psychobabble from the last section, we now need to talk about Web3.0 technologies to buy some semblance of legitimacy. The burgeoning metaverse, decentralized autonomous organizations (DAOs) leverage blockchain interoperability to facilitate cross-chain liquidity mining. Through non-fungible token (NFT) staking and decentralized finance (DeFi) yield farming, we enable permissionless smart contracts that optimize trustless consensus mechanisms. Our layer-2 scaling solutions utilize zero-knowledge proofs (ZKPs) to enhance cryptographic security, ensuring seamless integration with distributed ledger technologies (DLTs). Relax, the next section actually contains some information.

Conclusion

Enacting the Galaxy build process on a mobile device requires surprisingly little effort when using PostMarketOS (PmOS); a community effort to bring mainline Linux to smartphone devices, replacing the often vendor-locked operating systems (OS) that phones normally deploy with. PmOS provides not only full access to the smartphone's hardware and backend, but also comes with a choice of desktops. By cloning and compiling the Galaxy source, and installing the necessary dependencies via the Alpine *apk* package manager, Galaxy successfully built and hosted itself on an ARMv8 CPU without exceeding the phone's limited hardware capabilities. Current efforts to bring this to more ubiquitous phone OS's such as Android are dependent on the Termux package ecosystem which currently poses challenges during the build process.

UseGalaxy server and UI both running in ***Phosh*** *desktop via* ***PostmarketOS***

A Tmux session running Galaxy server and Htop

Firefox accessing the Galaxy UI on localhost

Alcatel Idol 3 4.7″
CPU Quadcore-1.2 GHz, ARMv8
RAM 1.3GB + 2GB swap
Chip Qualcomm MSM8916